World Wide Web & HTTP Protocol Suite

Coverage: HTTP/1.0 vs HTTP/1.1 (Persistent vs Non-Persistent, Pipelining), HTTP/2 (Binary Framing), HTTP/3 (QUIC), Cookies & Web Caching

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1. HTTP Message Architecture & Methods

- Stateless request-response protocol operating over TCP port 80 (HTTPS port 443).
- Methods: GET, POST, HEAD, PUT, DELETE. Status codes: 200 OK, 301 Moved, 304 Not Modified, 404 Not Found, 500 Error.
- State Management: Cookies (Set-Cookie / Cookie headers) and Session Tokens.

2. HTTP Protocol Evolution

- HTTP/1.0: Non-persistent (1 TCP handshake per object = 2 RTT per object).
- HTTP/1.1: Persistent connections by default; Pipelining (suffers Head-of-Line blocking).
- HTTP/2: Binary Framing Layer, Stream Multiplexing over single TCP connection, HPACK compression, Server Push.
- HTTP/3: Operates over UDP using QUIC transport layer to eliminate TCP Head-of-Line blocking and enable 0-RTT handshakes.

■ Worked Problem: Web Page Retrieval Time

- Given: 1 Base HTML + 8 JPEG images on same server. RTT = 40 ms.
- 1. Non-Persistent HTTP: 9 objects * 2 RTT = 18 RTT = 720 ms.
- 2. Persistent HTTP without pipelining: 1 RTT (TCP) + 1 RTT (Base) + 8 * 1 RTT = 10 RTT = 400 ms.
- 3. Persistent HTTP with pipelining: 1 RTT (TCP) + 1 RTT (Base) + 1 RTT (Images) = 3 RTT = 120 ms.

■ KTU University High-Yield Exam Questions

- Q: Compare Non-Persistent and Persistent HTTP connections with timing diagrams. (7 Marks)
- Q: Explain how HTTP/2 and HTTP/3 solve Head-of-Line blocking. (8 Marks)